Higher National Diploma in Engineering (Electrical and Electronics)

First Year, First Semester Examination — 2021

EE 1209 - Electronics Circuits and Systems

Question 1

(i) List the applications of diodes. (2 marks)

(ii) Explain the drift current for semiconductors. (3 marks)

(iii) Briefly explain the differences among an **Insulator**, **Conductor**, and **Semiconductor** with support of the energy band. (4 marks)

(iv) Explain the formation of a **PN junction diode**. (5 marks)

(v) Describe the PN junction diode's **forward** and **reverse bias characteristics** by referring necessary curve. (6 marks)

Question 2

(i) What is a capacitance diode? (2 marks)

(ii) Explain the requirement of a current limiting resistor for Zenor diode circuits. (3 marks)

(iii) Describe the avalanche effect. (4 marks)

(iv) Explain the operation of the full wave rectifier with relevant circuit, graphs. (5 marks)

(v) A full wave rectifier is used to energize 15 V d.c. to a pure resistive load having 1 kΩ and the diode has a forward resistance of 25 Ω. Estimate the peak value of the ac voltage required at the input side. (6 marks)

Question 3

(i) What is the lead-acid accumulator and its Ah? (2 marks)

(ii) What are various configurations associated with amplifier circuits? (3 marks)

(iii) Draw C filter and LC filter circuits and explain the operation. (4 marks)

(iv) a. Sketch the PNP and NPN transistor symbols. (2 marks)
b. Describes the operation of an NPN transistor. (3 marks)

(v) Determine the base, collector, and emitter currents and VCE for the common emitter amplifier circuit shown in Figure 3(v). Where VBE=0.7 V, β=150. (6 marks)

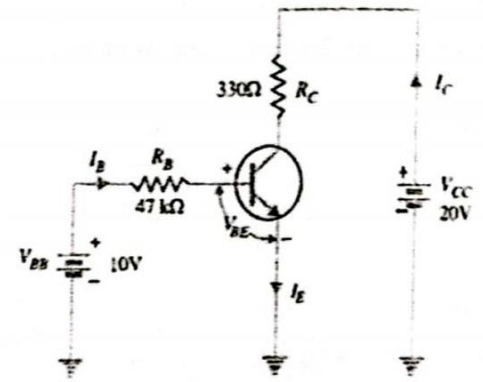

Figure.3 (v)

Question 4

(i) When can a transistor be used as a switch? (2 marks)

(ii) a. Why is 'FET' referred to as a "voltage-controlled device"? (2 marks)
b. Describe the operating principles of a junction field effect transistor (N-channel type). (3 marks)

(iii) Derive an expression for trans-conductance for JFET. (3 marks)

(iv) What is MOSFET and briefly explain N-channel depletion type MOSFET? (4 marks)

(v) A fixed biased JFET amplifier having parameters IDSS=20 mA & VGS(Off) =−8 V is shown in Figure 4(v). Determine VGS, ID, and VDS. (6 marks)

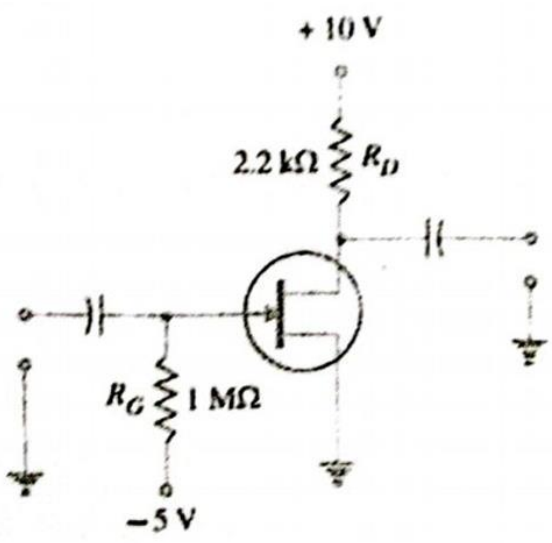

Figure 4. (v)

Question 5: Digital Logic and Number Systems

(i) Convert the following hexadecimal numbers to **binary**: a. $27BC.D8_{16}$ (1 marks) (ii) Convert the following octal number to a **hexadecimal number**: a. 1452_8 b. 367_8 (4 marks) (iii) a. Explain the rules of adding two **BCD numbers**. (2 marks) b. Add $0101 + 1001$ by using the BCD number. (3 marks) (iv) a. What is a **combinational logic circuit**? (2 marks) b. State why combinational logic circuits differ from the **sequential logic circuit**? (3 marks) (v) a. Find the **2's complement** and **1's complement** of 101101. (2 marks) b. Subtract 1000011_2 from 1010100_2 using **2's complement method** and show the **direct subtraction** for comparison. (3 marks)

Question 6: Boolean Algebra and Logic Gates

(i) State the **Commutative Property** and **Associative Property** of Boolean algebra. (3 marks) (ii) By using a **lamp and switches** draw an electrical equivalent circuit for: a. **NOT gate** b. **AND gate** c. **OR gate** (3 marks) (iii) States **de Morgan's theorems** for two logic variables A and B. (4 marks) (iv) A logic circuit shown in Figure 6(iv) consists of three inputs A, B, C, and the output is X . a. Write a **Boolean expression** for output b. Draw the **truth table** (4 marks) (v) Use a **Karnaugh map** to simplify the following: a. $X = AC' + A'B + ABC$ b. $F(w, x, y, z) = \Sigma(0, 1, 3, 7, 9, 10, 11, 12, 13)$ (6 marks)